

Total No. of Questions : 4]

SEAT No. :

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[6010] 155

B.E. (Honours in Virtual Reality and Augmented Reality) (Insem)

APPLICATION DEVELOPMENT USING AUGMENTED

REALITY AND VIRTUAL REALITY

(2019 Pattern) (Semester - VIII) (410703)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Make suitable assumptions if necessary.
- 4) Figures to the right indicate full marks.

Q1) a) What is Augmented Reality? How is Augmented Reality (AR) different from Virtual Reality (VR)? **[4]**

b) What technologies have been integrated into virtual reality? **[5]**

c) What do you understand by Vuforia SDK and how it can be used in AR? **[6]**

OR

Q2) a) What are the Applications of Augmented Reality? **[3]**

b) Which are the different components are involved in System architecture of Augmented reality application? **[6]**

c) How many AR toolkits available that can be integrated with Visual Studio for AR development? **[6]**

Q3) a) What is VRTK and why we use it? **[3]**

b) How can cameras be used in VR concept integration, and what are some examples of camera-based VR systems? **[6]**

c) What are some common challenges in debugging VR projects, and how can developers overcome them? **[6]**

OR

Q4) a) What is Google VR SDK, and what are its key features? **[4]**

b) What types of motion tracking systems are commonly used in VR, and what are their advantages and limitations? **[5]**

c) What are some best practices for optimizing text in VR interfaces, and why are they important? **[6]**

